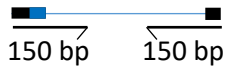


1. Quality check and trimming raw reads

paired-end Illumina sequencing



Up to 20 billion
read pairs



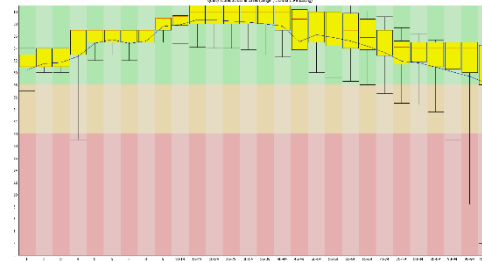
Forward
reads



Reverse
reads

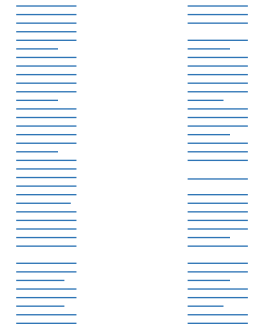


Fastqc quality plots



fastp

Trimmed and filtered
reads of good quality



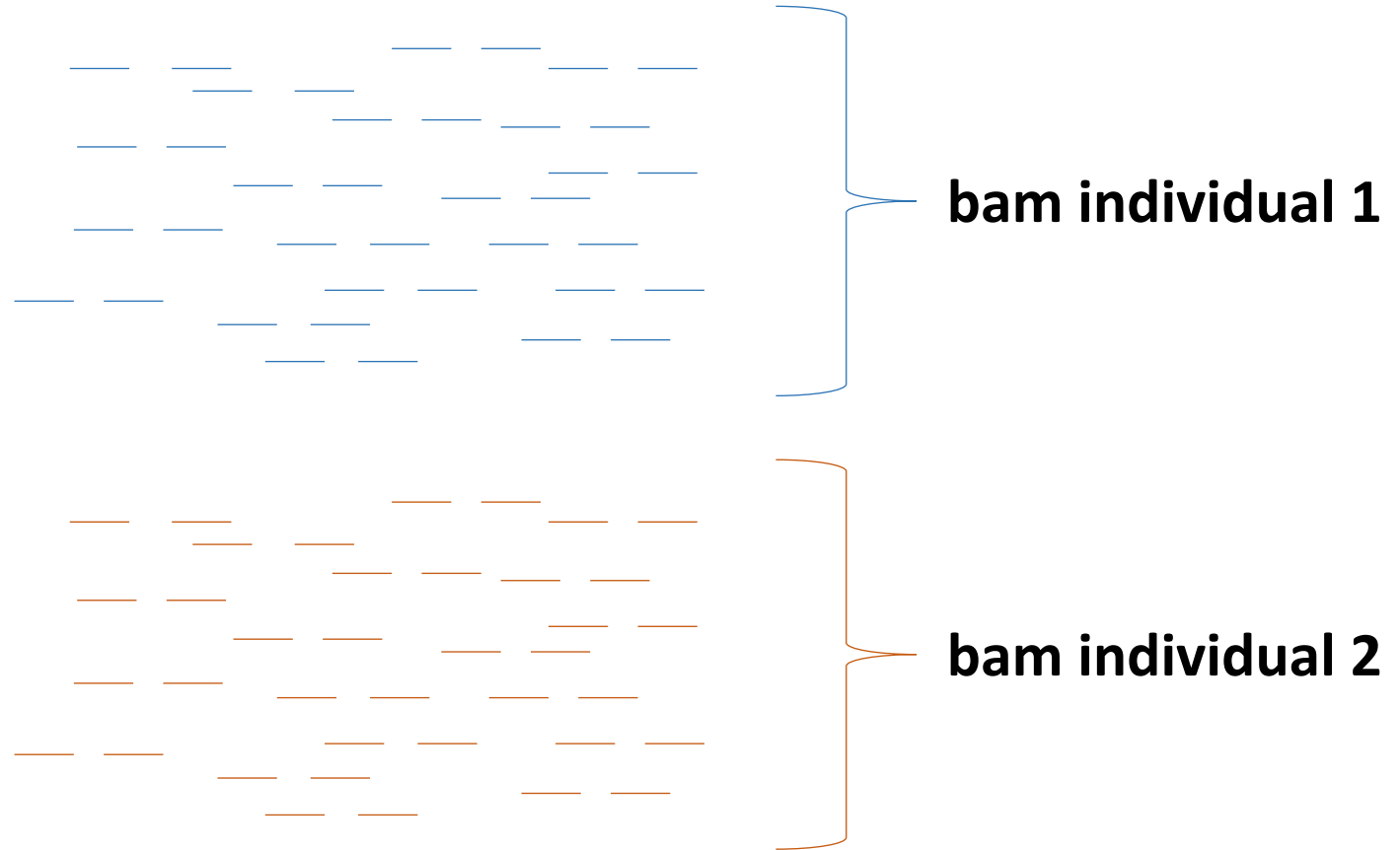
Fastq file of reads

```
@HWUSI-EAS611:34:6669YAAXX:1:1:5069:1159 1:N:0:
TCGATAATACCGTTTTTTTCCGTTTGATGTTGATACCAT
+
IIHIIHIIIIIIIIIIIIIIIIIIIIIIIIIIIIHIIIIHIIII
```

2. Alignment to the reference genome with bwa

reference

bwa



3. Variant and genotype calling with bcftools

reference

T

bcftools

T

T

T

T

C

C

C

C

vcf file

3. Variant and genotype calling with bcftools

reference

pos. 82154

Chromosome 1

bcftools

T

T

T

T

T

C

C

C

C

vcf file

vcf file: Genotypes for each individual at genomic sites

##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype call in the format REF/ALT"

#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT

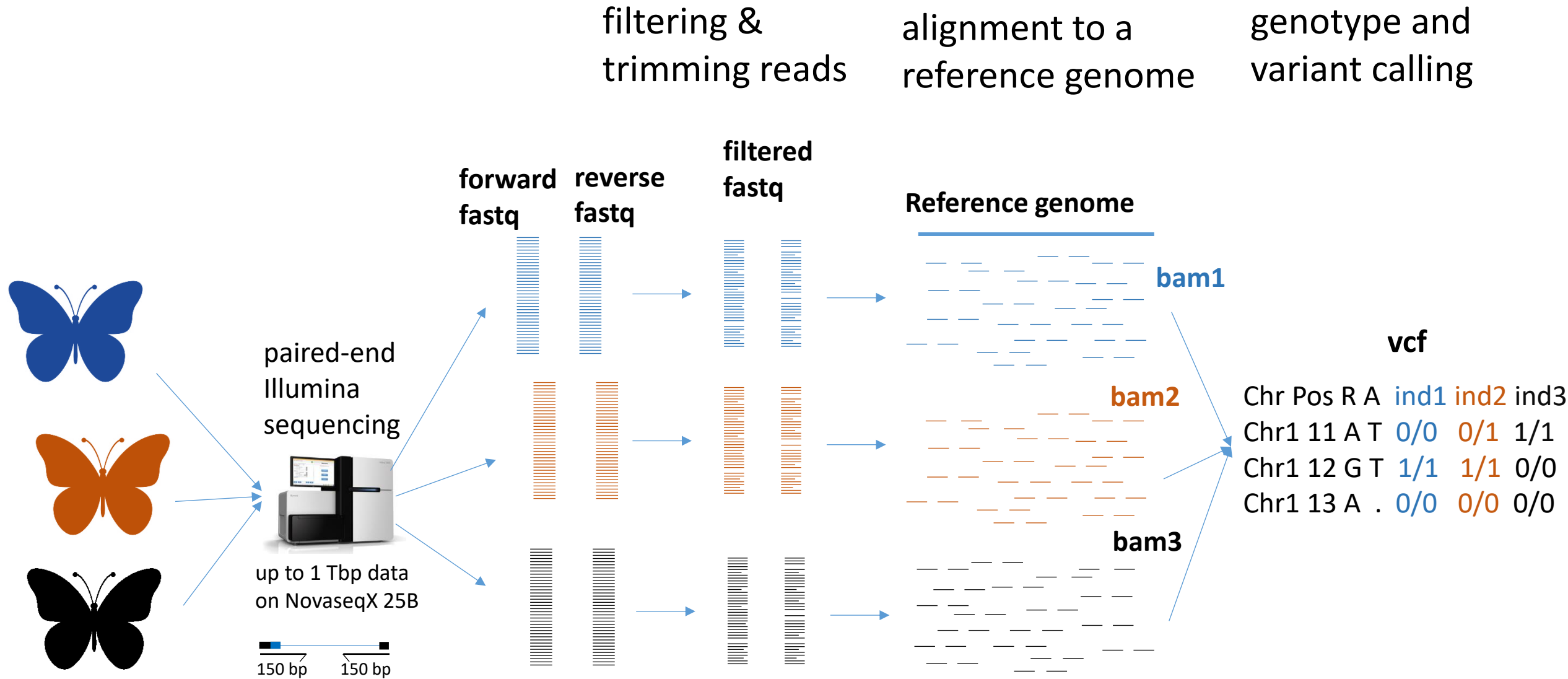
chr1 82154 . T C . GT 0/0 1/1 0/0 1/1 0/0

chr1 752566 . T . . GT 0/0 0/0 0/0 0/0 0/0

chr1 752721 . T C . GT 1/1 1/1 1/1 1/1 1/1

chr1 752721 . A . . GT ./. 0/0 0/0 0/0 0/0

A different visualisation of the same steps

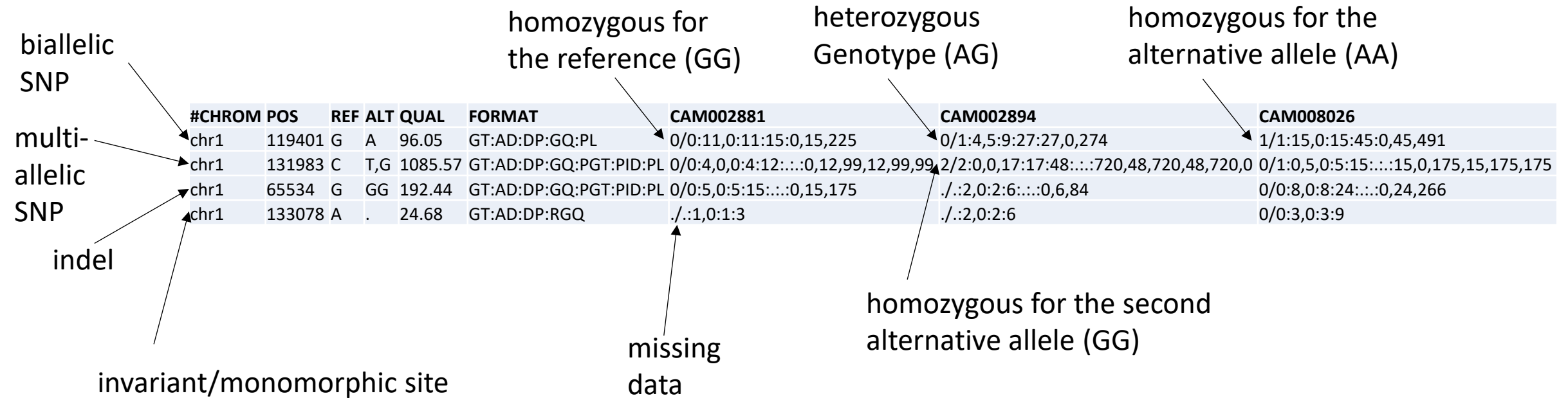


Vcf file structure

- For every sequenced position in the genome, there is a row
- For every individual, there is a column

```
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"  
#CHROM  POS  ID   REF  ALT  QUAL  FILTER  INFO    FORMAT  GEN  
chr1    82154  .    T    C    .    GT      0/0  1/1  0/0  1/1  0/0  1/1  
chr1    752566  .    T    .    .    GT      0/0  0/0  0/0  0/0  0/0  0/0  
chr1    752721  .    T    C    .    GT      1/1  1/1  1/1  1/1  1/1  1/1  
chr1    752728  .    A    .    .    GT      ./.  0/0  0/0  0/0  0/0  0/0
```

Vcf file format



The diagram illustrates the VCF file format with a table of four rows. Annotations with arrows point to specific parts of the table:

- biallelic SNP**: Points to the first row (chr1, 119401, G, A).
- multi-allelic SNP**: Points to the second row (chr1, 131983, C, T,G).
- indel**: Points to the third row (chr1, 65534, G, GG).
- invariant/monomorphic site**: Points to the fourth row (chr1, 133078, A, .).
- homozygous for the reference (GG)**: Points to the FORMAT field of the first row (GT:AD:DP:GQ:PL).
- heterozygous Genotype (AG)**: Points to the FORMAT field of the second row (GT:AD:DP:GQ:PGT:PID:PL).
- homozygous for the alternative allele (AA)**: Points to the FORMAT field of the third row (GT:AD:DP:GQ:PGT:PID:PL).
- missing data**: Points to the FORMAT field of the fourth row (GT:AD:DP:RGQ).
- homozygous for the second alternative allele (GG)**: Points to the FORMAT field of the second row (GT:AD:DP:GQ:PGT:PID:PL).

#CHROM	POS	REF	ALT	QUAL	FORMAT	CAM002881	CAM002894	CAM008026
chr1	119401	G	A	96.05	GT:AD:DP:GQ:PL	0/0:11,0:11:15:0,15,225	0/1:4,5:9:27:27,0,274	1/1:15,0:15:45:0,45,491
chr1	131983	C	T,G	1085.57	GT:AD:DP:GQ:PGT:PID:PL	0/0:4,0,0:4:12:....:0,12,99,12,99,99	2/2:0,0,17:17:48:....:720,48,720,48,720,0	0/1:0,5,0:5:15:....:15,0,175,15,175,175
chr1	65534	G	GG	192.44	GT:AD:DP:GQ:PGT:PID:PL	0/0:5,0:5:15:....:0,15,175	./.:2,0:2:6:....:0,6,84	0/0:8,0:8:24:....:0,24,266
chr1	133078	A	.	24.68	GT:AD:DP:RGQ	./.:1,0:1:3	./.:2,0:2:6	0/0:3,0:3:9

Next step: variant and genotype calling

Many tools available:

- Bcftools – fast, easy and good
- GATK – state of the art but only if the individuals to map are closely related to the reference genome (designed for human data)
- FreeBayes – also widely used
- ANGSD – for low-coverage data (<10x coverage)
- For RAD/GBS data:
 - STACKS
 - pyRAD
- For PacBio data:
 - DeepVariant (but only good for vertebrates or other organisms with low diversity)
 - bcftools: best for species like insects or plants with high genetic diversity

Filtering the vcf file

original vcf	filtering genotypes	filtering individuals	filtering sites
Chr Pos R A ind1 ind2 ind3 Chr1 11 A T 0/0 0/1 1/1 Chr1 12 G T 1/1 1/1 0/0 Chr1 13 A . 0/0 0/0 0/0	Chr Pos R A ind1 ind2 ind3 Chr1 11 A T 0/0 0/1 1/1 Chr1 12 G T ./. Chr1 13 A . 0/0 ./.	Chr Pos R A ind1 ind2 ind3 Chr1 11 A T 0/0 0/1 1/1 Chr1 12 G T ./. Chr1 13 A . 0/0 ./.	Chr Pos R A ind1 ind3 Chr1 11 A T 0/0 1/1 Chr1 12 G T ./. Chr1 13 A . 0/0 0/0
Lightly filtered vcf file	Remove genotypes with - too few reads - low genotype quality - uneven allele count	Remove individuals with - too much missing data - too low average sequencing depth - contamination/PCA outliers	Remove sites with - too much missing data - too high or low average sequencing depth - Note: sex chromosomes require separate filters
Additional filters for specific analyses (examples)	- more stringent filters - phasing	- subsampling for even coverage of populations - outgroups	- indels - monomorphic sites - repeat-masking - low minor allele frequency